Code No: 133AJ **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, March – 2022
DIGITAL LOGIC DESIGN
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
- - -

1.a) Convert the given Gray code number to equivalent binary 1001001011110010.
b) Convert $(A0F9.0EB)_{16}$ to decimal, binary, octal.
c) What is meant by 'min-term' and 'max-term'? Write the procedure to obtain the canonical SOP form of a given logic functions. [5+5+5]

2.a) Express the following functions in sum of min-terms and product of max-terms.
i) (xy+z) (y+xz) ii) B′D+A′D+BD.
b) Obtain the minimal SOP expression for the switching function using k-map.
Y= ∑ m(1,5,7,13,14,15,17,18,21,22,25,29) + ∑ d(6,9,19,23,30) draw and explain the logic diagram. [5+10]

3.a) Simplify the expression Y= π (7, 9, 10, 11, 12, 13, 14, 15) using the k-map method.
b) Draw the schematic diagram and truth table for half adder. Explain the design approach for half adder using universal gates. Draw the logic diagrams with relevant expressions. [7+8]

4.a) Design, draw and explain a 4-bit binary carry look ahead adder.
b) Describe the operations performed by the following logic circuits with an example:
(i) Comparator (ii) Decoder (iii) Encoder. [9+6]

5.a) Explain the operation of a 3-to-8 decoder 74LS138. Realize 4-to-16 decoder using two 3-to-8 decoders.
What is meant by 'edge triggered'? Differentiate SR-FF and JK-FF with their functional operation and excitation tables.
b) Draw and explain the circuit diagram of positive edge triggered J-K flip-flop using NOR gates with its truth table. How race around conditions are eliminated? [7+8]

6.a) Discuss about synchronous and ripple counters. Compare their merits and demerits.
b) What do you mean by universal shift register? Draw and explain its circuit diagram and operation. [7+8]

7.a) Using PLA logic, implement a BCD to excess-3 code converter. Draw and explain its truth table and logic diagram.
b) Explain in brief, about logic construction of 32×4 ROM. Draw and explain the relevant logic diagram. [7+8]

8.a) Derive the PLA programming table for the combinational circuit that squares a 3-bit number. Draw the relevant logic diagram.
b) A ROM chip of 4,096 × 8 bits has two chip select inputs and operates from a 5-volt power supply. How many pins are needed for the integrated circuit package? Draw and explain the relevant block diagram. [7+8]

---ooOoo---

JNTUH Question Papers March 2022